Computer Architecture
CS 325 - 001
Department of Physics and Computer Science
Medgar Evers College
Exam 2 - Make Up

Instructions:

- **Complete on May 25 by 10:00 pm.**
- **Use the IAS instruction set:**

Mnemonic	Opcode	Description
LMA	0A	Transfer contents from MQ to AC
LDM	09	Transfer M(X) to MQ
STA	21	Transfer contents from AC to memory location X
LDA	01	Transfer M(X) to AC
LDN	02	Transfer -M(X) to AC
ALD	03	Transfer |M(X)| to AC
ALN	04	Transfer -|M(X)| to AC
BRL	0D	Takes next instruction from left half of M(X)
BRR	0E	Takes next instruction from right half of M(X)
BPL	0F	If AC $>= 0$, takes next instruction from the left half of M(X)
BPR	10	If AC $>= 0$, takes next instruction from the right half of M(X)
ADD	05	Add M(X) to AC; put result in AC
AAD	07	Add |M(X)| to AC; put result in AC
SUB	06	Subtract M(X) from AC; put result in AC
ASB	08	Subtract |M(X)| from AC; put result in AC
MUL	0B	Multiply M(X) by MQ; put most significant bits of result in AC; least significant in MQ
DIV	0C	Divide AC by M(X); put quotient in MQ and remainder in AC
LSH	14	Multiply AC by 2
RSH	15	Divide AC by 2
STL	12	Transfer AC[28:39] to M(X)[8:19]
STR	13	Transfer AC[28:39] to M(X)[28:39]
HLT	00	Halts

- **Cheating of any kind is prohibited and will not be tolerated.**
- **Violating and/or failing to follow any of the rules will result in an automatic zero (0) for the exam.**

Grading

Section	Maximum Points	Points Earned
1	8	
Total	8	

1. **Given the program,**

```
000: 010AF0F003
001: 01061060D5
002: 0D00400000
003: 01061050D5
004: 210AF090AF
005: 0B0AF0A000
006: 210AF00000
```

perform an instruction cycle trace that lists all sequential changes to registers and memory starting from address 000, given the initial values:

Variable	Address	Initial Value
L	**AF**	**C64E4F3EF1**
M	**61**	**0000000050**
N	**D5**	**000000001E**

Recall branch command modifies the PC, but right command modifies IBR as well.

It should be 38 steps.